

Homogenized Planet Imagery

A Process-Art Critique of Space Image in the Age of AI

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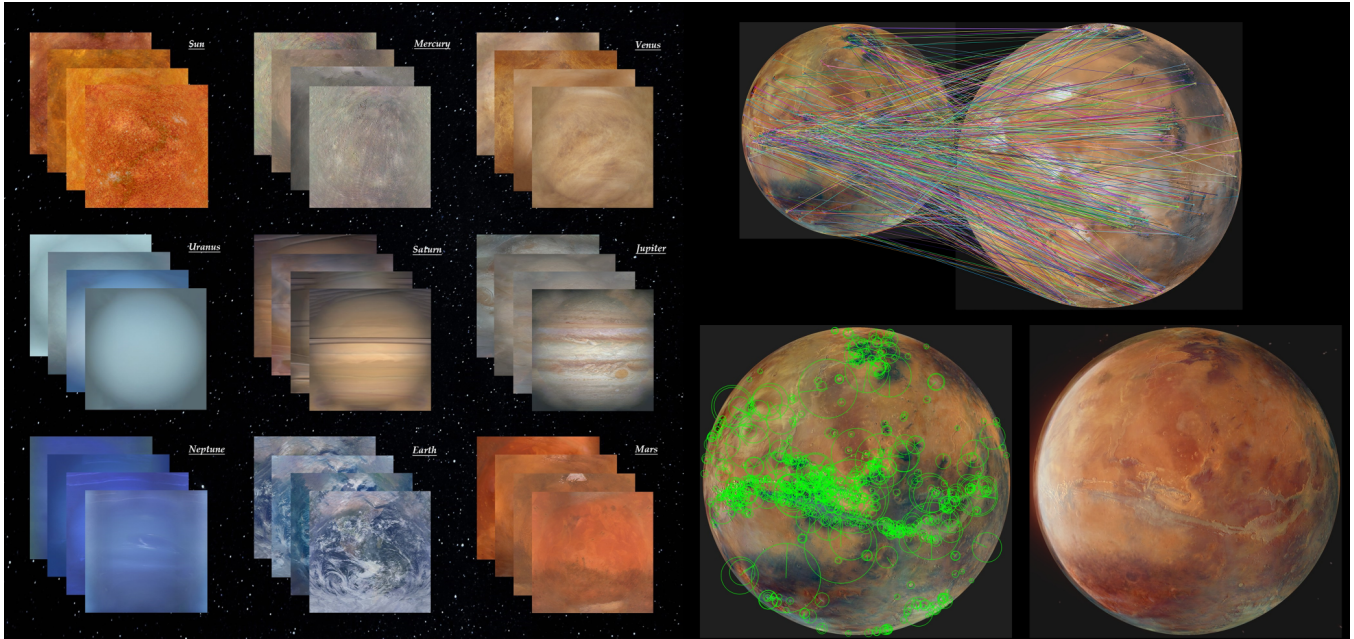


Figure 1: Collected images in *Homogenized Image Planet* and examples of image processing.

Abstract

In the paper we introduce Homogenized Image Planet, a multi-stage process artwork that examines how planetary images acquire authority today. We collect visuals from NASA archives, Wikipedia, search platforms, and generative AI; archive their meta-data; and fuse them with equal weights after feature-based alignment. The resulting texture animates a Unity and WebGL installation and is preserved on a radiation-tolerant chip launched to low-Earth orbit, forming a physical–virtual digital twin. Framed by power/knowledge, computational objectivity, and simulacra, audience surveys and expert interviews indicate decreased trust in

“official” imagery and increased critical awareness of algorithmic mediation.

CCS Concepts

• Applied computing → Media arts.

Keywords

Process Art, Planetary Imagery, Computational Objectivity, Simulacra

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1 Introduction: When We “Gaze at the Stars Through Images”

In the digital age, the public can hardly see distant planets with the naked eye. Intuitive ideas about how planets look now rest

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on four kinds of imagery: (1) scientifically calibrated images released by NASA and similar agencies, which carry technical authority and factual narratives; (2) illustrative figures in encyclopedias, which convey simplified information through community standards and workflows; (3) search-engine results, where ranking algorithms—shaped by commerce and popularity—govern visibility; and (4) images generated by AI models whose “hyperreal” look arises from training data biases and prompt design. These four poles compete as they shape what counts as “the real,” producing a multi-centered visual ecology rather than a single authority. This section sets out our key questions and contributions.

1.1 A Multi-Source Media Structure for Planetary Images

Planetary imagination today is molded by multiple sources: agency images, after layers of calibration and mosaicking, speak for “facts” and technical authority [Daston and Galison 2007; Galison 1999; Le Moigne et al. 2012]; encyclopedia figures, guided by community norms and layout templates, deliver proceduralized neutrality, while search ranking—driven by business aims and user profiles—decides what becomes most visible [Noble 2018]; finally, generative AI, steered by training data and prompts, produces new, self-standing visual surfaces that merely “look right” [Rombach et al. 2022]. Together these sources yield a competitive, multi-centered authority structure—from the iconic “Blue Marble” consensus [Reilly 2025], through the platform politics of visibility, to the hyperreal effects reproduced by generative models. In such a setting, the line between “real” and “rendered” is unsettled; as Baudrillard notes, when signs detach from their referents, they enter a self-maintaining hyperreal order [Baudrillard 1994]. We therefore move toward a procedural intervention and method test: by juxtaposing and rearranging sources, we observe shifts in audience perceptions of authority and critical awareness.

1.2 Project Overview: Framework and Aims

Homogenized Image Planet is a process-as-artwork study that juxtaposes and “equalizes” multi-source planetary images to expose how visual authority is made, translating theory into an operational framework. Using four inputs (space agencies, encyclopedias, search platforms, and generative AI), the project runs through four linked phases: collecting and recording sources and metadata; balancing and re-stating visual differences; staging a two-site viewing context; and evaluating public perception and judgment (implementation details follow). The project centers on homogenizing “authoritative” images together with visuals from multiple sources and applying them to a virtual “planet” installation. A second key implementation involves collaborating with a space art program and a commercial space company to embed the homogenized images and the virtual “planet” digital twin on a chip carried into low-Earth orbit, thereby closing the loop in practice. To date, the chip continues to orbit the Earth. Conceptually, the work brings together the distribution and re-distribution of power and knowledge [Foucault 1980], the sign logic of the hyperreal [Baudrillard 1994], and the “digital twin” idea of cross-checking virtual and physical contexts [Dahmen and Rossmann 2021]. Empirically, questionnaires and interviews track shifts in trust and critical stance,

while process archives support verifiability and reuse. The goal is to expose the construction of image authority with minimal prior assumptions, setting a baseline for testable dialogue between art practice and public science.

1.3 Objectives and Contributions

Our workflow begins with a practical structure, from collecting images (NASA/Wiki/Search/AI) and storing metadata, to homogenizing textures via OpenCV + SURF, and finally staging both a Unity→WebGL installation and a flash memory chip carried to orbit—then test the effect of juxtaposition: through audience surveys and expert interviews, we assess how co-presenting homogenized visuals with “real space” contexts shifts judgments along the real and simulacral axis. Contributions are threefold. Conceptually, we propose “homogenization”: equalizing the weights of NASA, Wikipedia, search, and AI imagery to challenge hierarchical ways of knowing planets [Baudrillard 1994; Steyerl 2025]. Methodologically, we define a four-stage process-art paradigm—image collecting (disentangling power/knowledge) → homogenization (simulacral re-production) → virtual installation (digital-twin arena) → in-orbit counterpoint (physical-image juxtaposition)—with the process itself as artwork. Practically, we archive the homogenized texture and the virtual installation on a radiation-tolerant chip that rides a commercial launch to orbit, completing an experiment that puts “virtual image vs. actual cosmos” in the same domain.

2 A Single Theoretical Thread and Its Method Interface

We argue along one continuous line: starting with Foucault’s power/knowledge, to explain who and which workflows/platforms decide “what counts as a real image”; moving to Daston & Galison’s mechanical and computational objectivity, treating calibration, registration, and normalization in scientific imaging as mechanisms that can be repurposed; and closing with Baudrillard’s simulacra/hyperreal, showing how equal-weight juxtaposition produces a sign surface that stands in for the referent. The four-stage workflow makes this line operational and measurable [Baudrillard 1994; Daston and Galison 2007; Foucault 1980; Galison 1999].

2.1 How Image Authority Is Made: From Institutions to Platforms (Foucault)

Across production and circulation, the sequence “selection—annotation—release” draws the visible boundary of “facts”: space agencies fix a version of the “real” via scientific calibration and mosaicking; encyclopedia communities proceduralize “neutrality” through editing norms and templates; commercial search and social platforms decide visibility through ranking and recommendation [Foucault 1980; Noble 2018]. Authority is not a given; it is built by institutional workflows and platform infrastructures. Our first stage therefore collects images by source–time–processing chain and surfaces metadata, turning the power chain into an object of analysis rather than a silent backdrop [Crawford and Paglen 2021; Noble 2018].

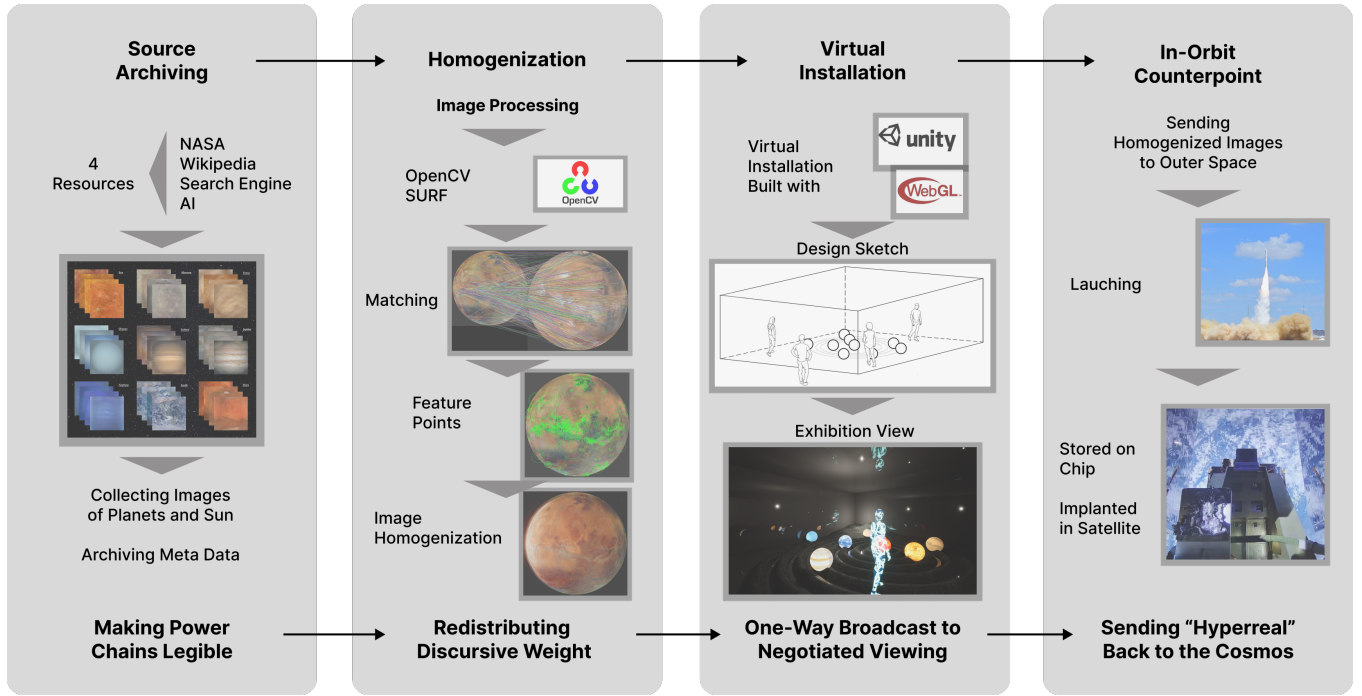


Figure 2: The core four-stage workflow, technical steps, and how they align with the conceptual framework.

2.2 Mechanizing “Objectivity”: From Mechanical to Computational (Daston/Galison)

Scientific images are not naturally “unbiased.” Historically, objectivity shifts from idealized depiction to mechanical objectivity, to computational objectivity that relies on procedures for repeatability [Daston and Galison 2007; Galison 1999]. Steps such as calibration (radiometric/color), registration (geometry/projection), and normalization (photometric/statistical) form a technical ethic that eats away differences to produce sameness. We do not claim to reproduce scientific imaging; instead, we convert this mechanism into a critical tool: a transparent chain that shows how “neutral-looking” computation rewrites visibility at the pixel level and provides a testable interface for equal-weight juxtaposition (details in §3).

2.3 Simulacra and the Hyperreal (Baudrillard)

When the visible force of an image no longer depends on how closely it matches the referent, but on a “look-effect” optimized by workflows and platforms, it enters an order that is “more real than the real” [Baudrillard 1994]. After equal-weight fusion erases source differences, viewers face not “which image is truer,” but a single surface without provenance or hierarchy, whose persuasive power comes precisely from systematic handling of difference. This helps explain why, in our study, confidence in any single “official” image drops: homogenization and juxtaposition are technical operations and, at the same time, semiotic interventions into the criteria of the real (see §5–§6).

3 Critical Workflow: A Four-Stage Homogenization Framework

Homogenized Image Planet is not a single algorithm demo but a continuous chain from data collection to in-orbit counterpoint. We outline the four stages, keeping technical choices aligned with critical aims.

3.1 Source Archiving: Making Power Chains Legible

We gather images from four typical channels: (1) download multi-spectral, calibrated agency photos via NASA portals and APIs and collect their metadata [Le Moigne et al. 2012]; (2) retrieve images embedded in Wikipedia entries; (3) use a custom search engine to collect the top 100 hits for terms like “planet surface,” labeling source tendencies under the lens that “algorithm = discourse power” [Noble 2018]; and (4) generate 200 “hyperreal” candidates with prompts such as “satellite photo of Mars” [Rombach et al. 2022]. All visuals are stored with source, capture time, color space, and annotation type, explicitly revealing the power/knowledge chain [Foucault 1980].

3.2 Homogenization: Redistributing Discursive Weight at the Pixel Level

From each source we select 20 representative images. We extract keypoints with SURF and use RANSAC to curb mismatches, estimating homographies to align inputs [Bay et al. 2008]. We then assign equal weights (0.25) to the four sources and compute a pixel-wise average in a gamma-corrected linear space [Bradski 2000]. This

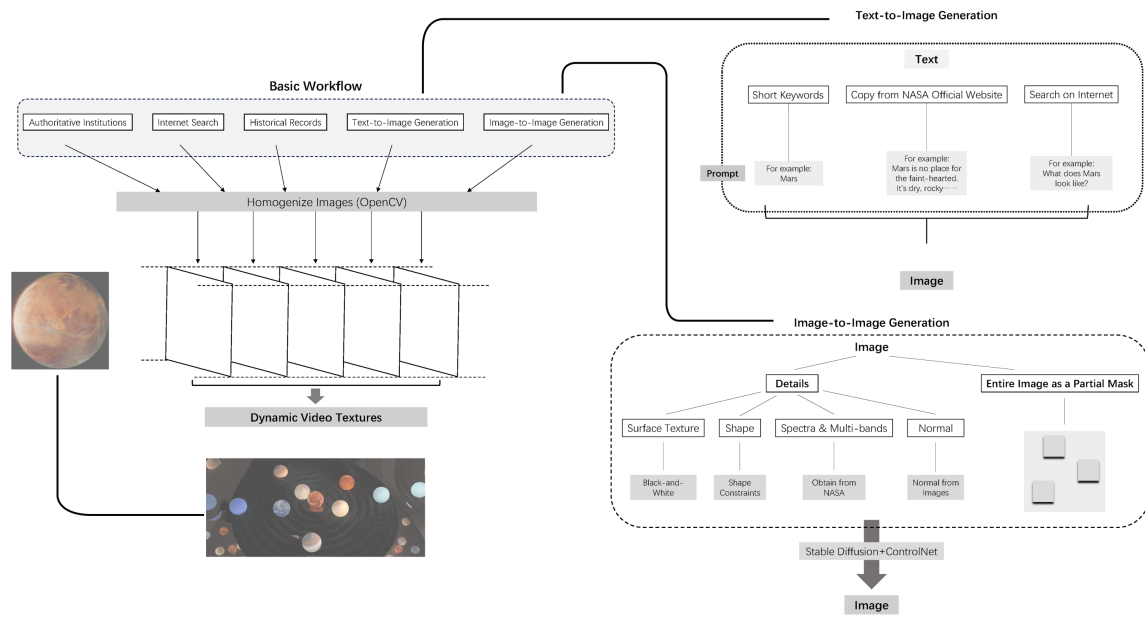


Figure 3: Image-processing pipeline for the collected imagery, including homogenized-texture generation and an outline of how generative AI is used.

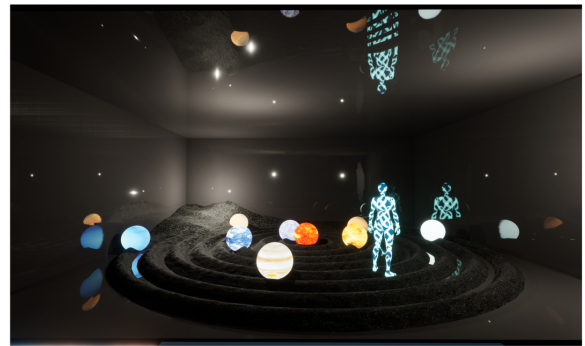
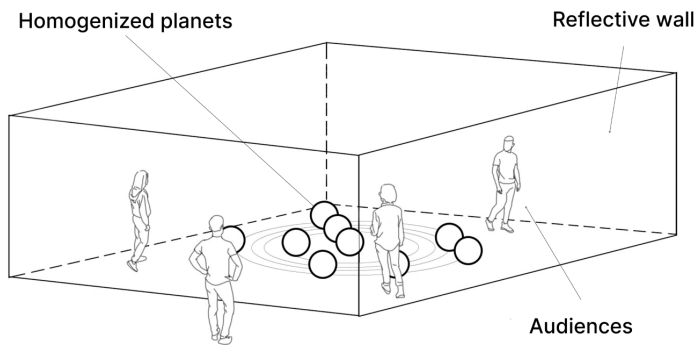


Figure 4: Design sketch and renderings of the virtual installation, with selected design details.

deliberate “even split” flattens NASA’s scientific code, community consensus, search-rank bias, and AI training bias into one visual surface—echoing Baudrillard’s claim that once multiple references are assimilated, the result tilts toward the hyperreal [Baudrillard 1994]. We also keep metadata indices in the output header to enable tracing, answering Crawford and Paglen’s critique of “black-box labels” in datasets [Crawford and Paglen 2021].

3.3 Virtual Installation: From One-Way Broadcast to Negotiated Viewing

In Unity we build a virtual installation featuring nine evenly spaced fictional planets. As visitors roam, surface textures shift through homogenized variants; the surrounding space uses black reflective materials so planets mirror back—an echo of Steyerl’s warning

about the self-replication of “power images” [Steyerl 2025]. To make provenance legible, we offered guided tours and made the provenance materials publicly available; this support enabled visitors to identify the images’ sources and understand how the subsequent homogenization was challenged and handled. For smooth WebGL performance, textures are compressed and lighting is baked.

3.4 In-Orbit Counterpoint: Sending the “Hyperreal” Back to the Cosmos

At the end of the chain, the homogenized texture and the virtual installation are written to a radiation-tolerant flash chip mounted on a low-Earth-orbit satellite. The mixed image enters outer space; the virtual and the celestial share the same domain. This closes the

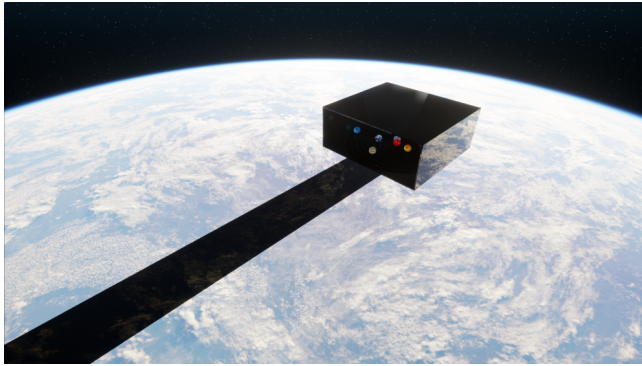


Figure 5: In the exhibition, the virtual installation is presented on a simulated near-Earth-orbit platform in outer space.

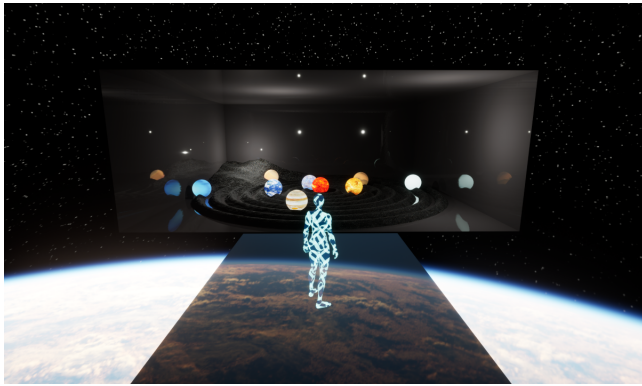


Figure 6: WebGL view of the virtual installation, where visitors can freely walk and explore the *Homogenized Image Planet*.

loop and throws our questions back at the work itself [Dahmen and Rossmann 2021].

4 Virtual Encounters: Exhibition and Launch in Tandem

4.1 A Brief Context for Space Art

Since the Moon Museum micro-tile rode Apollo 12 in 1969, space art has been a laboratory for critiquing the present. In 1977, Galloway and Rabinowitz staged Satellite Arts Project using NASA links for transcontinental performance [Galloway and Rabinowitz 1977]; in 2012, Trevor Paglen's *The Last Pictures* etched 100 images onto EchoStar XVI in GEO, a "post-human monument" [Paglen 2012]; the Moon Gallery (2022–2025) aims to install a micro-museum on the lunar surface [Moon Gallery Foundation 2025]. Unlike curator-led projects, a space-art DAO proposed a decentralized exhibition in 2022. The initial plan: (1) a metaverse warm-up to explain a "time-capsule" concept; (2) an in-orbit counterpoint by writing open-call and invited works to a radiation-tolerant chip for launch; and (3) later, hybrid physical and immersive layers. As invited artists, we designed *Homogenized Image Planet* to balance feasibility with

critique—roamable in browsers, traceable and questionable online; archivable, comparable, and re-interpretable in space.

4.2 Collaboration: DAO Deliberation and Project Delivery

From the outset the "space exhibition" rejected a single-venue model: key steps—scene layout, contract writing, on-chain minting—required community votes, institutionalizing Foucault's boundary of power/knowledge [Foucault 1980]. A 24-hour Discord sprint set early requirements; a virtual-architecture team built the metaverse shell. When the Unity scene first ran, browser constraints surfaced. We baked lighting, disabled real-time shadows, compressed materials, and built a digital twin of the installation [Dahmen and Rossmann 2021].

The final design—a futuristic open hall evoking outer space—met community expectations. During the online show, our installation went live on the DAO-hosted site [Space Art DAO 2022]. Afterward, an on-chain vote passed with 94.6% in favor to package 1 GB of homogenized textures and installation files into a radiation-tolerant flash chip [Irom and Nguyen 2008]. At year's end, the chip rode a commercial rocket [19, 20] to low-Earth orbit. As an aside, the DAO's role mirrors our aim: decentralization helps unsettle fixed links between authority and source.

5 Evaluation: Survey and Interview Design

To see whether the project truly unsettles intuitive boundaries between "real" and "simulacral," we ran both a quantitative survey and qualitative interviews. The survey probed shifts in trust and critical stance; the interviews invited six experts from art, technology, and aerospace to reflect on the meaning of "virtual-in-orbit juxtaposition," resonating with Foucault's interweaving of knowledge and power and with Steyerl's critique of "power images" shaping perception [Foucault 1980; Steyerl 2025]. More than 40 online visitors to the installation (abbr. 60% with art and tech backgrounds, the rest general audience) completed a 10-minute questionnaire including items adapted from media-credibility scales, indicators inspired by Watson-Glaser for critical thinking, and a short presence subscale (ITC-SOPI) for immersion. Six expert interviews covered: the import of virtual and real juxtaposition for art and technology; how provenance cues affect boundary judgments; contrasts between online installations and traditional in-person shows; and directions for improvement. We present representative quotes alongside quantitative results in the next section.

6 Discussion: Efficacy of Process Art and Its Political Charge

Results show strengths and limits. Quantitatively, a clear majority reported lower confidence in the exclusive "realness" of any one source after the experience, and many noted a stronger impulse toward critical reflection when provenance cues appeared. This dual effect fits Baudrillard's account of the hyperreal—when multiple signs stand side by side, old trust orders shake [Baudrillard 1994]—and aligns with Foucault's call to re-distribute power/knowledge [Foucault 1980], as well as with the historical move from mechanical to computational objectivity in scientific imaging [Daston and Galison 2007; Galison 1999]. Interviews also surfaced tensions between

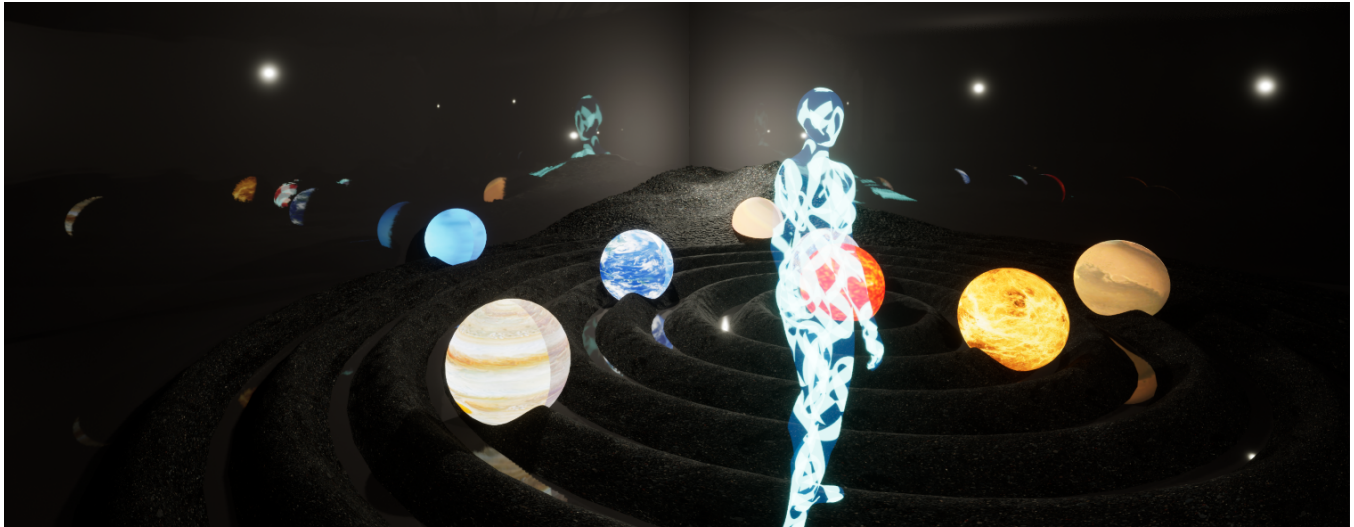


Figure 7: View of visitors closely observing the planets showing homogenized-image in the virtual space.

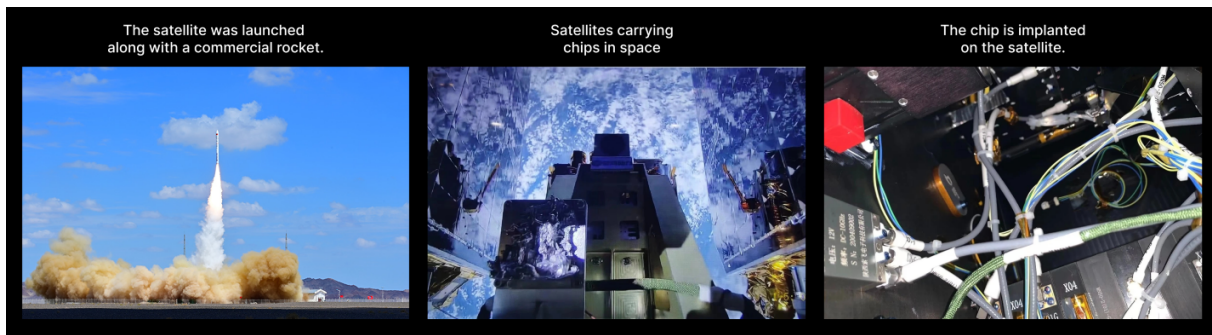


Figure 8: Launch and in-orbit documentation: the chip was delivered to low-Earth orbit together with the satellite. © Galantic Energy.

process art and physical installations. Curators praised the low barrier and reach of the online format; artists and engineers worried that browser-level experiences may lack the bodily impact of physical setups, risking a “click-through” critique. On re-allocating discursive power, the source visualization asks “who defines the real,” letting images compete in one arena. One expert remarked: “When NASA’s weight is only 25%, the ‘scientific real’ we reverberate turns out to be divisible.” This echoes Crawford and Paglen’s call to make bias visible [Crawford and Paglen 2021]. Others warned that too much flattening can erase professional value, suggesting a balance between equity and expertise in future versions. Coupling an in-orbit archive with a virtual show is quickly absorbed by the logics of commercial space and digital art—Paglen already warned that space romanticism risks being re-framed by new institutional contexts [Paglen 2012]. In short, the work lowers trust in “official real,” activates critical awareness, and demonstrates process art as a workable critical method—while also revealing fresh constraints. Future work (e.g., multi-source interpolation and in-orbit telemetry) aims to extend the chain, keeping viewers, scientists, and systems moving back and forth between the real and the hyperreal.

7 Limitations and Future Work

Although this project offers a first run at multi-source homogenization and a virtual-in-orbit pairing, several constraints remain. Technically, fusion can behave unstably or randomly on real-world inputs, and some images are ill-suited to a single pipeline; on the presentation side, browser limits on shading, textures, and memory cap the installation’s expressiveness. We will therefore tighten OpenCV registration and fusion parameters and refactor WebGL resource scheduling and rendering to improve stability and portability. A second limitation concerns interaction depth. Viewers cannot yet make continuous, controllable transitions across the four sources, nor access live orbital context while watching. Our next iteration will add shader-based interpolation and mixing controls, with a simplified HUD to organize provenance and status cues; the interface will draw on Turkle’s discussion of “conversational technologies” to reduce cognitive load and improve legibility [Turkle 2015]. Third, the digital-twin relation is not yet two-way. The current radiation-tolerant chip functions as a one-way time capsule and does not return parameters or state. We will explore

payload and link designs with telemetry to close the loop between virtual and physical domains, aligning more closely with Dahmen and Rossmann's notion of a bidirectional twin [Dahmen and Rossmann 2021]. Finally, data coverage and diversity need to grow. Our textures focus on the nine planets; comets, asteroids, and exoplanets are not yet included, and spectral variety is limited. We plan to incorporate multispectral and radar data and to adopt recent methods in multi/hyperspectral fusion, following Nalepa, to raise both scientific credibility and informational richness [Nalepa 2021].

8 CONCLUSION

Public understanding of the cosmos is now thoroughly mediated; NASA releases, encyclopedia communities, platform ranking, and AI generation together form the structure of visual authority. Our four-stage process—collection, homogenization, virtual staging, and in-orbit counterpoint—turns Baudrillard and Foucault into practice and uses the digital-twin idea to hold virtual and space contexts together. Evidence shows a meaningful drop in trust for “official” images and a rise in critical reflection, with experts recognizing both theoretical depth and technical invention. The project keeps its theoretical promise and offers a template for process art across digital art and public science. As capital and technology redraw the boundaries of critique, we will keep pushing multi-source interpolation, orbital return links, and public science tooling so that “the real” is launched, questioned, and remade again and again.

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References

- Jean Baudrillard. 1994. *Simulacra and Simulation*. University of Michigan Press, Ann Arbor, MI, USA. Translated by Sheila Faria Glaser; The Body, in Theory.
- Herbert Bay, Andreas Ess, Tinne Tuytelaars, and Luc Van Gool. 2008. Speeded-Up Robust Features (SURF). *Computer Vision and Image Understanding* 110, 3 (2008), 346–359. doi:10.1016/j.cviu.2007.09.014
- Gary Bradski. 2000. The OpenCV Library. *Dr. Dobbs's Journal of Software Tools* 25, 11 (Nov. 2000), 120–126.
- Kate Crawford and Trevor Paglen. 2021. Excavating AI: The Politics of Images in Machine Learning Training Sets. *AI & Society* 36 (2021), 1105–1116. doi:10.1007/s00146-021-01162-8
- Ulrich Dahmen and Jürgen Rossmann. 2021. What is a Digital Twin – A Mediation Approach. In *Proceedings of the 2021 IEEE International Conference on Electro Information Technology (EIT)*. IEEE, Mt. Pleasant, MI, USA, 165–172. doi:10.1109/EIT51626.2021.9491883
- Lorraine Daston and Peter L. Galison. 2007. *Objectivity*. Zone Books, New York, NY, USA.
- Michel Foucault. 1980. *Power/Knowledge: Selected Interviews and Other Writings, 1972–1977*. Pantheon Books, New York, NY, USA. Translated by Colin Gordon, Leo Marshall, John Mepham, and Kate Soper.
- Peter Galison. 1999. Reflections on Image and Logic: A Material Culture of Microphysics. *Perspectives on Science* 7, 2 (06 1999), 255–284. arXiv:https://direct.mit.edu/posc/article-pdf/7/2/255/1790427/posc.1999.7.2.255.pdf doi:10.1162/posc.1999.7.2.255
- Kit Galloway and Sherrie Rabinowitz. 1977. Satellite Arts Project (A Space with No Geographical Boundaries). Live telecollaborative performance using NASA satellite links. <https://www.ecafe.com/getty/SA/index.html> Produced by Mobile Image; real-time multi-site performance over satellite.
- Farokh Irom and Duc N. Nguyen. 2008. *Radiation Tests of Highly Scaled High Density Commercial Nonvolatile Flash Memories*. Technical Report JPL Publication 08-27. Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA. https://npp.nasa.gov/files/16425/08_168_1%20JPL_Farokh.pdf
- Jacqueline Le Moigne, Thomas G. Grubb, and Barbara C. Milner. 2012. NASA IM-AGESEER: NASA IMAGES for Science, Education, Experimentation and Research. In *Proceedings of SPIE 2012: Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XVIII*. International Society for Optical Engineering, Baltimore, MD, USA. NASA Goddard Space Flight Center, Document ID: 20120011804.
- Moon Gallery Foundation. 2025. *Moon Gallery: The First Permanent Extraterrestrial Art Gallery*. <https://moongallery.eu/> Mission targeting a 2025 lunar delivery.
- Jakub Nalepa. 2021. Recent Advances in Multi- and Hyperspectral Image Analysis. *Sensors* 21, 18 (2021). doi:10.3390/s21186002
- Safiya Umoja Noble. 2018. *Algorithms of Oppression: How Search Engines Reinforce Racism*. NYU Press, New York, NY, USA. doi:10.2307/j.ctt1pw9w5
- Trevor Paglen. 2012. *The Last Pictures*. University of California Press, Berkeley, CA, USA. Co-published with Creative Time Books (New York).
- Angus Reilly. 2025. *The power of the Blue Marble*. Engelsberg Ideas. <https://engelsbergideas.com/notebook/the-power-of-the-blue-marble/> Notebook.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, New Orleans, LA, USA, 10684–10695. https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR2022_paper.html
- Space Art DAO. 2022. *Cosmic Ephemeroptera*. <https://ce.spaceartdao.com/> Online exhibition microsite; project credited as “Powered by Space Art DAO” and described as sealed in a CERES-1 artwork-carrying satellite.
- Hito Steyerl. 2025. *Medium Hot: Images in the Age of Heat*. Verso, London; New York.
- Sherry Turkle. 2015. *Reclaiming Conversation: The Power of Talk in a Digital Age*. Penguin Press, New York, NY, USA.